Supplementary Information for
FALCON: Fast Algorithm for Large-scale Cross-domain Compositional Network Inference

Contents

S1 Derivation of the cross-domain centering identity

For independently normalised compositions $\mathbf{X} \in \Delta^{p-1}$ and $\mathbf{Y} \in \Delta^{q-1}$, let W_i^X, W_j^Y denote the underlying absolute abundances with totals $S^X = \sum_k W_k^X$ and $S^Y = \sum_l W_l^Y$. After the centered log-ratio transform,

$$\mathrm{clr}_X(i) \;=\; \log W_i^X - \tfrac{1}{p}\sum_{k=1}^{p} \log W_k^X \quad \text{(and analogously for } Y\text{)}. \tag{S1}$$

Define the true cross-covariance of the log-abundances, $\Omega_{ij} \;\triangleq\; \mathrm{Cov}(\log W_i^X, \log W_j^Y)$. Substituting and using bilinearity of covariance,

$$\begin{aligned} T_{ij} &\triangleq \mathrm{Cov}(\mathrm{clr}_X(i),\, \mathrm{clr}_Y(j)) \\ &= \Omega_{ij} - \tfrac{1}{p}\sum_k \Omega_{kj} - \tfrac{1}{q}\sum_l \Omega_{il} + \tfrac{1}{pq}\sum_{k,l} \Omega_{kl}. \end{aligned} \tag{S2}$$

The four terms have a compact matrix interpretation: each is a different component of a double-centered Ω.

Proposition 1 (Matrix form of the centering bias)**.** *Let $H_p = I_p - \frac{1}{p}\mathbf{1}_p\mathbf{1}_p^\top$ denote the p-dimensional centering matrix. Then $T = H_p\,\Omega\,H_q^\top$.*

Proof. $(H_p \Omega H_q^\top)_{ij} = \Omega_{ij} - \frac{1}{p}\sum_k \Omega_{kj} - \frac{1}{q}\sum_l \Omega_{il} + \frac{1}{pq}\sum_{k,l}\Omega_{kl}$, which matches (S2) term-by-term. The two outer sums expand to the marginal and double-marginal corrections; the inner product gives the bilinear term. □

Lemma 1 (Non-identifiability without regularisation)**.** *For any constants $a \in \mathbb{R}^p$ and $b \in \mathbb{R}^q$, $T(\Omega + \mathbf{1}_p a^\top + b\,\mathbf{1}_q^\top) = T(\Omega)$.*

Proof. $H_p \mathbf{1}_p = 0$ and $H_q^\top \mathbf{1}_q = 0$ because the all-ones vector lies in the null space of any centering matrix. Hence $H_p(\mathbf{1}_p a^\top)H_q^\top = 0$, and the same for the b shift. □

Lemma 1 formalises the central identifiability obstacle: the $(p+q-1)$-dimensional null space of $(H_p, H_q^\top)$ leaves Ω underdetermined. The ℓ_1 penalty in equation (7) of the main text breaks this degeneracy by selecting the unique solution of minimum mass among the infinite family compatible with T, encoding the biological prior that most phage–bacteria pairs are non-interacting.

S2 FISTA convergence and step size

Let $f(\Omega) = \frac{1}{2}\|T - H_p \Omega H_q^\top\|_F^2 + \frac{\lambda_2}{2}\|\Omega - \Omega_{\text{prior}}\|_F^2$. Using the idempotence $H_p^2 = H_p$ and symmetry $H_p = H_p^\top$,

$$\nabla f(\Omega) = H_p\,(H_p \Omega H_q^\top - T)\,H_q + \lambda_2(\Omega - \Omega_{\text{prior}}). \tag{S3}$$

Proposition 2 (Lipschitz constant)**.** *∇f is Lipschitz with constant $L \le 1 + \lambda_2$.*

Proof. The spectral norm of any centering matrix satisfies $\|H_p\|_2 = 1$ because its eigenvalues lie in $\{0,1\}$. Hence the linear map $\Omega \mapsto H_p \Omega H_q^\top$ has operator norm bounded by $\|H_p\|_2\|H_q\|_2 = 1$, and the gradient of the Frobenius residual is 1-Lipschitz. Adding the quadratic Tikhonov term contributes λ_2. □

We use step size $\eta = 1/L = 1/(1+\lambda_2)$. With Nesterov momentum [1, 2], FISTA achieves $F(\Omega_k) - F(\Omega^\star) = O(1/k^2)$, where $F(\Omega) = f(\Omega) + \lambda_1\|\Omega\|_1$. In the simulations of Section 3.5 of the main text, the relative change $\|\Omega_{k+1} - \Omega_k\|_F / \|\Omega_k\|_F$ drops below 10^{-4} within 20–30 iterations.

S3 Adaptive regularisation parameters

The sparsity parameter λ_1 controls the trade-off between recall and precision. We default to the BIC-style choice

$$\lambda_1 \;=\; \kappa \cdot \hat{\sigma}_T \sqrt{\log(pq)/n}, \tag{S4}$$

where $\hat{\sigma}_T$ is the standard deviation of the entries of the observed T matrix and κ is a sparsity prior in $(0,1)$ controlling how aggressively to threshold. The scaling $\sqrt{\log(pq)/n}$ matches the near-optimal rate for sparse signal detection in regularised regression. Practitioners with stability-based criteria (e.g. StARS [3]) may replace this default by repeatedly subsampling the data and choosing the largest λ_1 at which the empirical edge variability is below a nominal $\beta = 0.05$.

The prior weight λ_2 is typically small (10^{-2} to 10^{-1} times $\|T\|_F^2/(pq)$) when biological priors are partial or uncertain. Setting $\lambda_2 = 0$ disables the prior entirely.

S4 Multiplicative replacement and the CLR bias

The CLR transform requires strictly positive entries. We follow [4]:

$$x_i^{\text{rep}} = \begin{cases} \delta & \text{if } x_i = 0, \\ x_i \cdot (1 - n_0\delta) & \text{otherwise,} \end{cases} \qquad \delta = \frac{0.65}{p^2}, \tag{S5}$$

where n_0 is the number of zero entries in the row. This preserves the unit sum constraint exactly. The residual bias introduced by zero replacement scales as $O(1/p^2)$ in the CLR coordinates (cf. Lemma 3 of [5]) and is therefore negligible in the regimes we target ($p \geq 50$).

S5 Random projection: dimension selection

RandProp uses the Johnson–Lindenstrauss lemma [6, 7] to embed the column-normalised CLR matrix $\tilde{Z} \in \mathbb{R}^{n \times p}$ into a d-dimensional sketch. For pairwise inner-product preservation within tolerance ε with probability $\geq 1 - 1/p^2$, we set

$$d = \left\lceil 8 \ln(p)/\varepsilon^2 \right\rceil, \tag{S6}$$

clipped to $\min(d, n)$ since exceeding the sample dimension yields no additional information. The Achlioptas sparse construction draws each entry of the projection matrix from $\{+1/\sqrt{d/3}, 0, -1/\sqrt{d/3}\}$ with probabilities $\{1/6, 2/3, 1/6\}$, reducing the effective multiplication count by a factor of three without sacrificing the JL guarantee.

The top-k extraction step uses `numpy.argpartition` (introselect) to obtain the k largest absolute correlations per row in $O(p)$ per row. Refinement recomputes the exact ρ_p for the candidate set in $O(nk)$ per row.

S6 Extended simulation tables

Table S1: Single-domain detection power at $p = 100$ (mean $\pm$ standard deviation across 20 replicates). Power is the fraction of true edges detected at BH-FDR < 0.05 with a minimum effect-size filter of $|\rho_p| > 0.1$.

$\rho \setminus n$	50	100	200	500
0.2	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
0.3	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.58 ± 0.09
0.4	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.99 ± 0.01
0.5	0.00 ± 0.00	0.00 ± 0.00	0.01 ± 0.02	1.00 ± 0.00
0.7	0.00 ± 0.00	0.00 ± 0.00	0.66 ± 0.05	1.00 ± 0.00

S7 Reproducibility checklist

- Algorithms: `src/falcon/__init__.py` (Python 3.10+, NumPy $\geq$ 1.24, SciPy $\geq$ 1.10).
- Benchmark runner: `benchmarks/run_on_server.py` with `multiprocessing.Pool`, regenerating every CSV table in `data/` (`scalability.csv`, `detection.csv`, `cross_domain.csv`, `fdr_control.csv`) from documented random seeds.

Table S2: Single-domain detection power at $p = 500$. The conservative behaviour reflects BH-FDR adjustment over $\binom{500}{2} \approx 1.25 \times 10^5$ tests, not an intrinsic failure of the proportionality estimator: pairwise $|\rho_p|$ estimates remain accurate (Pearson correlation with truth > 0.95 in all conditions; not shown). RandProp's top-k pre-selection or stability-based thresholding [3] is required at this scale.

$\rho \setminus n$	50	100	200	500
0.2	0.00	0.00	0.00	0.00
0.3	0.00	0.00	0.00	0.00
0.4	0.00	0.00	0.00	0.00
0.5	0.00	0.00	0.00	0.00
0.7	0.00	0.00	0.00	0.00

Table S3: Cross-domain inference at $p = 60$, $q = 80$, $n = 300$, 20 planted interactions (70% lytic / negative), 20 replicates. Significance threshold $|C| > 0.1$ and Fisher-z $p < 0.05$.

Method	Correlation	Sign acc.	Sensitivity	Specificity	Bias
Naive CLR	0.993 ± 0.005	1.000	1.000	0.9489 ± 0.004	0.0006
SparXCC-like	0.993 ± 0.005	1.000	1.000	0.9489 ± 0.004	-0.00005
CrossNet	0.993 ± 0.005	1.000	1.000	$\mathbf{0.9622 \pm 0.003}$	-0.00009

- CSV I/O helpers: `benchmarks/io_utils.py`.
- Figure generation: `manuscript/figures/generate_workflow.py`, `generate_single_domain.py`, `generate_cross_domain.py`. Each script reads only the CSV tables in `data/` so figures can be regenerated without re-running the benchmarks.
- Repository will be archived on Zenodo at the time of acceptance with a permanent DOI; see Data and Code Availability.

References

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Table S4: Wall-clock benchmarks on a single CPU core (Apple Silicon, NumPy 1.26 with Accelerate BLAS), median of 3 runs.

n	p	FastProp (s)	RandProp (s)
100	100	0.0002	0.007
100	500	0.002	0.022
100	1000	0.008	0.045
100	2000	0.040	0.157
100	5000	0.315	0.366
500	500	0.003	0.061
500	1000	0.010	0.117
500	2000	0.043	0.259
500	5000	0.420	1.696
1000	2000	0.052	0.697

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